An easy yet useful definition of optical depth (τ) is by the formula $I = I_0 e^{-\tau}$, where I is irradiance, and I_0 is the initial irradiance. For our particular data, we have optical depths at altitudes 15-20, 20-25, 25-30, and 30-35 km. For ease of notation, let us call these τ_1, τ_2, τ_3, and τ_4. Given an amount of insolation I_0, as it passes through the 30-35 km range, the irradiance becomes $I_0 e^{-\tau_4}$. This remaining irradiance now passes through the 25-30 km range, becoming $I_0 e^{-\tau_4} e^{-\tau_3}$ Continuing this pattern, the final irradiance after passing through the stratosphere is $I_0 e^{-\tau_1} e^{-\tau_2} e^{-\tau_3} e^{-\tau_4} = I_0 e^{-(\tau_1+\tau_2+\tau_3+\tau_4)}$. Therefore, the optical depth from radiation passing through the entire stratosphere (15-35 km) can be described as the sum of the optical depths of each 5 km layer.